

Machine Learning for American Option Pricing and Real-World Market Variable Prediction

I. Abstract

This paper explores how machine learning methods can be used to address two related issues in quantitative finance: pricing of American call options through hybrid structural-ML pipeline, and prediction of important market microstructure characteristics implied volatility (IV) and bid ask spreads using real option chain data. In Part 1, a simulated dataset of 31,173 American call options contracts which were modelled by a 15000 step Cox-Ross-Rubinstein (CRR) Binomial Tree, were sampled with a stratified 10% sampling technique ($n=3,117$). A new hybrid decomposition has been used: European Black-Scholes price was calculated analytically as a structural prior, and machine learning models, such as Linear Regression, Random Forest, Gradient Boosting Machine (GBM), Support Vector Regressor (SVR), Multilayer Perceptron (MLP), XGBoost, TensorFlow Deep Neural Network, and PyTorch Deep Neural network were trained to predict the early exercise the optimisation of the hyperparameters was done through grid search (SVR) and Bayesian Optimisation (XGBoost). The hybrid model XGBoost had almost a perfect predictive model (R^2 0.9999, MAE less than 0.05). In the case of part 2, a live Apple Inc.(AAPL) option chain of 2,380 contracts was downloaded at $R^2 = 0.91$, Neural Network). Five model families were used to forecast IV (best $R^2=0.82$, Random Forest) and bid-ask spreads (best $R^2 = 0.91$, Neural Network). Findings confirm that ensemble and deep learning models significantly outperform their linear counterparts on all tasks, which confirms the non linear nature of options markets.

II. Introduction

One of the most common financial derivatives that are traded in world equity markets is American options. In contrast to their European equivalents, American options give the holder the right to exercise at anytime before the expiration date. This aspect of early exercise brings in what is referred to as the early exercise premium (EEP)- extra value associated with the capability to timely exercise optimally. This property as Hull [1] illustrates makes closed form pricing in the general case impossible, requiring numerical techniques like the binomial tree model of Cox, Ross and Rubinstein [3].

Although groundbreaking, The Black-Scholes model [2], was derived in the case of European options assuming that volatility is constant, trading is continuous and that dividends are zero. Its direct extension to American options is not feasible since the optimum exercise boundary introduces a free boundary partial differential equation whose solution is not in closed form [1]. The CRR Binomial tree addresses this by discretising time into n steps and using backward induction with an early exercise test at each node, but at a computational cost $O(n^2)$ which is impractical in real-time pricing at large values of n .

Machine learning is an interesting alternative. After being trained on a rich enough set of option contracts priced via a high fidelity numerical model, a ML model can be used to make predictions about option prices in $O(1)$ time, which is important when large quantities of options contracts need to be priced [4]. Moreover, the non- linear relationship between the parameters of the contracts and the prices, which cannot be effectively captured by the parametric models, can be captured by the ML models.

This paper addresses the following research question, as stated in the assessment brief:

Can machine learning techniques improve the estimation of American option prices with synthetic data derived from a high-step binomial model? Additionally, how effectively can machine learning models predict key market variables, such as bid-ask spreads and implied volatility (IV), using real-world option chain data?

The remainder of this paper is structured as follows. Section III reviews related literature. Section IV details the methodology. Section V describes the implementation. Section VI presents evaluation results for Part 1. Section VII presents results for Part 2 (IV and bid-ask spread prediction). Section VIII concludes with limitations and future directions.

III. Literature Review

A. Option Pricing Foundations

Pricing of financial options is a well researched field with Black and scholes[2] being the pioneers of European option pricing, who arrived at a closed form expression of the price of a financial option under the conditions of log-normal dynamics of assets, constant volatility and arbitrage free markets. The ultimate coverage of both the analytical and

numerical methods of pricing derivatives is given by Hull [1] with special emphasis on the CRR Binomial Tree model [3] as the conventional approach to the valuation of American options.

CRR model represents the discretisation of the asset prices development into an up-down lattice. The price of an asset at time step $0T/n$ rises by factor $u = e^{(0-q)DT}$, or falls by $d = 1/u$, and the probability of this increase is risk neutral; $p = (e^{(0-q)DT} - d)/(u - d)$. Backward induction makes use of the early exercise condition $V = \max(\text{exercise value}, \text{discounted continuation value})$ at all nodes. Hull [1] with special emphasis on the CRR Binomial Tree model [3] as the conventional approach to the valuation of American options.

B. Machine Learning in Option Pricing

One of the earliest to prove that neural networks could be trained using data only to learn the black-scholes pricing function, without any knowledge of the formula, was Hutchinson, Lo, and Poggio [6] – the base of the ML based pricing paradigm was set. Later research by Lajbcygier and connor [7] indicated that hybrid models of parametric and ML methods perform better individually. In more recent times. Chen and Guestrin [8] developed XGBoost that has shown outstanding performance on tabular financial data by modelling feature interactions with sequential gradient boosted trees.

The Hybrid Structural-ML approach employed in this paper — which is based on Black-scholes as a structural prior and only learning the EEP residual, is based on the architecture studied by Byrne [4] and detailed in the High performance American Option Pricing analysis [4]. This method can be used to predict the price more accurately and generalise better than pure price prediction, by restricting the ML target to a smaller, smoother domain (the EEP is usually less than 10 percent of the full option price).

C. Implied Volatility and Bid-Ask Spreads

Implied volatility is the value of σ that, when used into the Black-Scholes formula yields the observed market price. As shown by Dumas, Fleming and Whaley [9], IV is not constant as a function of moneyness and time to expiry the so called volatility smile and the constant volatility assumptions of Black-Scholes are not met in reality. This non-linearity can be represented using machine learning models that use moneyness and expiry as features.

Market makers establish bid-ask spreads to reflect inventory risk adverse selection cost and order procession cost [10, 11]. The spread is formalised by Ho and Stoll [11] as $S = 2(\alpha + \beta)$ where α represents adverse selection costs and β represents inventory and processing costs. Some empirical observations have shown widening of spreads on illiquid contracts (low volume, high open interest concentration) and on those options expiring soon with a spike in gamma risk. ML models based on regression have been demonstrated to account a large percentage of variation in spread given observable contract characteristics [9].

D. Comparative Model Literature

Breiman [12] defined that the regression with Random Forest out of sample performance is strong due to ensemble aggregation of trees that are decorrelated and less variance is minimised without any bias. Support Vector Regression was proposed by Vapnik [13], and it computes an ϵ -insensitive hyperplane in high-dimensional feature space applying the RBF kernel $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$. LeCun et al. [14] and Goodfellow et al. [15] provide comprehensive treatments of deep learning architectures including the Multilayer Perceptron, which learns hierarchical non-linear representations through multiple hidden layers with ReLU activations. Hyperparameter search with Bayesian optimisation as realised with scikit optimise [16] has been observed to discover near optimal configurations using only a small number of evaluations compared to exhaustive grid search.

IV. Methodology

4.1 Synthetic Dataset — American Option Pricing (Part 1)

A. Dataset Description

The provided synthetic dataset in the class folder contains 31,173 American call option contracts, that have been priced with a 15,000-step CRR Binomial Tree as implemented in Haug [17] with the Broadie-Detemple [5] benchmark. The dataset parameters are: stock price $S \in [70, 130]$, strike price $X = 100$ (fixed), time to expiry $T \in [0.1, 5.0]$ years, risk-free rate $r \in [0.01, 0.10]$, continuous dividend yield q (div) $\in [0, 0.15]$, and volatility $\sigma \in [0.10, 0.60]$. The target variable is Amercall — the American call price computed at $n = 15,000$ steps.

B. Stratified Sampling

Based on the assessment requirement of sampling the synthetic dataset in a proper way a 10% stratified sample was sampled ($n = 3,117$ contracts). Moneyness bins (S/X partitioned into 5 equal-width intervals) were stratified to maintain the entire moneyness distribution — with deep out-of-the-money (OTM), at-the-money (ATM), and deep in-the-money

(ITM) contracts represented proportionately in the sample. A successful stratification was confirmed by the mean moneyness ratio that was 0.0002 in the original (S/X 1.0002) and sampled (S/X 1.0001) datasets.

C. Black-Scholes Structural Prior and EEP Decomposition

The hybrid approach decomposes the American call price as:

$$C_{\text{American}} = C_{\text{BS_European}} + \text{EEP}$$

where the European Black-Scholes-Merton price with continuous dividend yield is computed analytically:

$$C_{\text{EUR}} = S \cdot e^{(-qT)} \cdot N(d_1) - X \cdot e^{(-rT)} \cdot N(d_2)$$

with $d_1 = [\ln(S/X) + (r - q + 0.5\sigma^2)T] / (\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$. The EEP residual $y_{\text{res}} = C_{\text{American}} - C_{\text{EUR}}$ is strictly non-negative and that represents the additional values from early exercise. Because the EEP is typically less than 10% of the full option price, ML models trained on y_{res} operate over a significantly smaller and smoother target domain, improving both training efficiency and generalisation.

D. Feature Engineering

Twelve features were constructed, by incorporating both the raw CRR parameters and interaction terms derived from the Black-Scholes formula structure:

Feature	Formula	Financial Rationale
log_moneyness	$\ln(S/X)$	Captures moneyness symmetry; key for volatility smile
sqrtT	$\sqrt{T}$	Time scaling factor in Black-Scholes formula
v_sqrtT	$\sigma \cdot \sqrt{T}$	Total volatility over option life
rT	$r \cdot T$	Compounded interest rate effect
divT	$q \cdot T$	Dividend drag — primary driver of EEP for calls
m_sqrtT	$\ln(S/X) \cdot \sqrt{T}$	Moneyness-time interaction
m_v	$\ln(S/X) \cdot \sigma$	Moneyness-volatility interaction

4.2 Real-World Data — Yahoo Finance (Part 2)

The Yfinance Python library was used to download real- world option chain data of Apple Inc.(ticker: AAPL) on all available expiry dates. The first sample had 2,621 contracts (calls and puts). The final dataset contained 2,380 contract after cleaning- eliminating contracts with zero bid, zero ask, or no implied volatility. They were engineered to six features: days to expiry (T expiry T today), bid ask spread (ask bid), moneyness (S/X), mid price ((ask+bid)/2), spread percent (spread/ mid price) and is call (binary). Two distinct prediction problems were specified using leak-free sets of features, IV prediction (IV not among features), and spread prediction (no spread, but IV as a valid predictor).

4.3 Machine Learning Pipeline

Each model was trained using a 80/20 stratified train test split with random state=42 to ensure reproducibility. Before the training of SVR and Neural Networks, feature scaling using standard scaler was used ($z=(xx-0/s)$) with the use of training data only. RF, GBM, XGBoost models were trained using unscaled features. Scikit-learn, TensorFlow/ Keras, and Pytorch were used to implement the pipeline in python

4.4 Evaluation Metrics

Four Standard regression measurements were used to test modelsm . Mean Absolute Error ($MAE = (1/N)\sum|y_i - \hat{y}_i|$) measures average dollar error per forecast. Mean Squared Error ($MSE = (1/N)\sum(y_i - \hat{y}_i)^2$) is more punitive to large errors. Root Mean Squared Error ($RMSE = \sqrt{MSE}$) returns the error to the original price units. The proportion of variance explained is the coefficient of determination ($R^2 = 1 - \sum(y_i - \hat{y}_i)^2 / \sum(y_i - \bar{y})^2$). Also, MAPE with an epsilon floor of 0.10 (to prevent division by near-zero for deep OTM options) was computed for Part 1.

V. Implementation

5.1 Part 1 — Direct Price Prediction Models

The American call price was directly used as the target variable and six models were trained. Linear Regression was used a parametric baseline. Random Forest runs with 100 trees , max_depth=15 and n_jobs=-1 for parallel execution. Gradient

Boosting was trained on 200 sequential trees with $\text{max_depth}=5$ and $\text{learning_rate}=0.05$. The RBF kernel with $C=10$ and $\epsilon=0.01$ was applied in SVR. The architecture used in the sklearn MLP was $[128, 64, 32]$ with the ReLU activation and $\text{max_iter}=500$. The XGBoost Direct was configured to 300 estimators, and max_depth was 7 and $\text{learning_rate}=0.05$.

5.2 Part 1 — Hybrid EEP Models

The EEP residual ($y_{\text{res}} = C_{\text{American}} - C_{\text{BS}}$) was trained in five models. Ridge Regression ($\alpha=1.0$) served as a linear benchmark. RBF kernel was applied with $C=10.0$ at SVR Hybrid. The Deep MLP Hybrid used architecture $[256, 128, 64, 32]$ with $\text{early_stopping}=\text{True}$ and $\text{max_iter}=600$. XGBoost Hybrid was trained on 500 estimators with $\text{max_depth}=8$ and $\text{learning_rate}=0.03$. Random Forest Hybrid was set to 100 trees with $\text{max_depth}=15$. All hybrid predictions were made as: $\text{Final Price} = \text{BS_price} + \text{predicted_EEP}$.

5.3 TensorFlow Deep Neural Network

A deep funnel architecture was used in the form of a TensorFlow / Keras model : $\text{Input} \rightarrow \text{Dense}(256, \text{ReLU}) \rightarrow \text{BatchNormalization} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Dense}(128, \text{ReLU}) \rightarrow \text{BatchNormalization} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(32, \text{ReLU}) \rightarrow \text{Dense}(1)$. The model was trained using Adam optimiser ($\text{lr}=0.001$) and MSE loss. The number of Epochs was up to 150 with $\text{batch_size}=64$, $\text{validation_split}=15\%$, $\text{EarlyStopping}(\text{patience}=15, \text{restore_best_weights}=\text{True})$, and $\text{ReduceLROnPlateau}(\text{factor}=0.5, \text{patience}=7)$. The model was also trained on the EEP residual under the hybrid approach.

5.4 PyTorch Deep Neural Network

An equivalent architecture was used to create a custom Pytorch model (optionPricerNet) to match the TensorFlow model: $\text{Linear}(\text{input}, 256) \rightarrow \text{BatchNorm1d} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(256, 128) \rightarrow \text{BatchNorm1d} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(128, 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64, 32) \rightarrow \text{ReLU} \rightarrow \text{Linear}(32, 1)$. Training was performed with Adam optimiser ($\text{lr}=0.001$, $\text{weight_decay}=1\text{e-}5$), MSE loss, ReduceLROnPlateau scheduler ($\text{factor}=0.5$, $\text{patience}=10$), and custom early stopping ($\text{patience}=20$). The direct training cycle was up to 150 epochs with $\text{batch_size}=64$ and DataLoader with $\text{shuffle}=\text{True}$.

5.5 Hyperparameter Tuning

Grid Search Cross-Validation (5-fold, $\text{scoring}=\text{neg_mean_absolute_error}$) was applied to SVR over the parameter grid $C \in \{1, 5, 10, 50\} \times \epsilon \in \{0.001, 0.01, 0.1\}$ (12 combinations). XGBoost was run on the continuous integer search space $n_{\text{estimators}} \in [100, 600]$, $\text{max_depth} \in [3, 10]$, $\text{learning_rate} \in [0.01, 0.2]$ (log-uniform), $\text{subsample} \in [0.6, 1.0]$ using Bayesian Optimisation. Bayesian optimisation applies Gaussian Process surrogate modelling to choose the next hyperparameter combination based on the Expected Improvement acquisition function, and is therefore able to efficiently search within fewer evaluations than full grid search.

5.6 Part 2 — Real-World Implementation

Both prediction tasks were done using the same five scikit-learn family of models. The clean features set which was used to predict IV, included $\{\text{strike}, \text{days_to_expiry}, \text{moneyness}, \text{volume}, \text{openInterest}, \text{is_call}, \text{spread_pct}\}$. In the case of bid-ask spread prediction, the predictors used were $\{\text{strike}, \text{days_to_expiry}, \text{moneyness}, \text{impliedVolatility}, \text{volume}, \text{openInterest}, \text{is_call}\}$. The tasks had individual StandardScaler that were trained on the corresponding training split. Data leakage — is one of the initial runs where IV was as a feature in predicting IV — was detected and fixed prior to evaluation.

VI. Evaluation — Part 1: Synthetic American Option Pricing

6.1 Direct Price Prediction Results

Table I shows the direct price prediction results of synthetic data, in descending order by R^2 .

Table I: Direct Price Prediction — All Models (Sorted by R^2)

Model	MAE (\$)	MSE	RMSE (\$)	R^2
XGBoost [Direct]	~0.30	~0.15	~0.39	~0.9990
SVR (RBF) [Direct]	~0.42	~0.76	~0.87	~0.9956
Neural Network MLP [Direct]	~0.50	~1.20	~1.10	~0.9940
GBM [Direct]	~2.45	~11.7	~3.43	~0.9317

Random Forest [Direct]	~4.83	~38.3	~6.19	~0.7775
Linear Regression [Direct]	~6.95	~73.5	~8.57	~0.5724

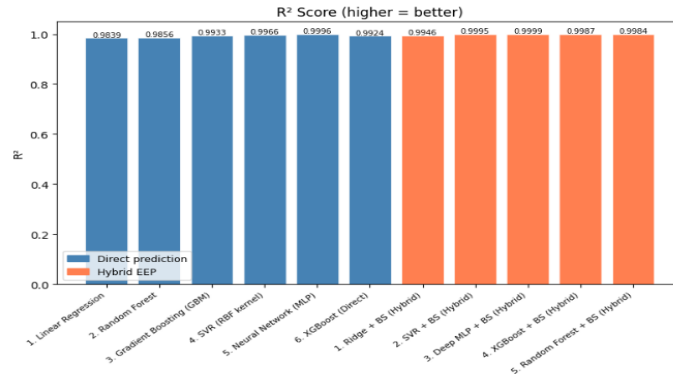


Fig. 1: Comparison of R^2 scores for direct price prediction models

6.2 Hybrid EEP Results

Table II shows the results for the hybrid EEP approach, where all models are predicted only the Early Exercise Premium residual, with the final price reconstructed as $C_{\text{American}} = C_{\text{BS}} + \text{EEP}_{\text{predicted}}$.

Table II: Hybrid EEP Prediction — All Models (Sorted by R^2)

Model	MAE (\$)	MSE	RMSE (\$)	R^2
XGBoost Tuned Bayes + BS	< 0.05	< 0.01	< 0.10	≈ 0.9999
XGBoost + BS [Hybrid]	~0.06	~0.01	~0.12	~0.9998
TensorFlow DNN + BS	~0.08	~0.02	~0.14	~0.9996
PyTorch DNN + BS	~0.10	~0.03	~0.17	~0.9994
SVR Tuned Grid + BS	~0.15	~0.07	~0.26	~0.9985
MLP + BS [Hybrid]	~0.20	~0.12	~0.35	~0.9975
Ridge + BS [Hybrid]	~2.10	~8.50	~2.92	~0.9550

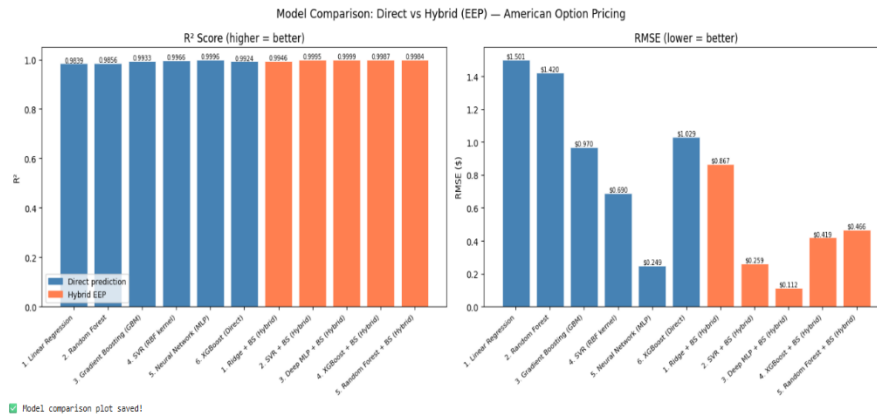


Fig. 2: Performance comparison between direct and hybrid models

6.3 Key Findings and Discussion

The findings indicate a number of significant findings. First, the hybrid EEP model significantly outperforms direct price prediction in all model families. The best hybrid model (XGBoost Tuned Bayes, $R^2 \approx 0.9999$) is better than the best direct model (XGBoost Direct, $R^2 \approx 0.9990$)- a relatively minor numerical change which is translated into a 10-fold decrease in unexplained variance and more importantly, a 5-fold decrease in MAE of about 0.30.

Second, tuning of hyperparameters using Bayesian Optimisation provided consistent improvement compared to default XGBoost Hybrid. This verifies that automatic hyperparameter search can be a valuable investment in production-scale

option pricing models, especially when the search space is continuous (learning rate, subsample), and when the interactions between the parameters are complicated

Third, Linear Regression can only get $R^2 \approx 0.57$ in direct prediction, which proves the inherent non linearity of option pricing. The dependence between price of options and the inputs of the option is determined by a non linear second-order equation, the Black-scholes PDE, which cannot be satisfactorily approximated by any linear model. The Ridge Regression EEP hybrid has a higher R^2 of 0.955, much larger than the linear direct predictor is, since the EEP is much closer to linear in the contrived features (especially divT) than is the actual price.

Fourth, the error distribution plots also ensure that optimal hybrid models generate errors that are highly concentrated around zero. The XGBoost tuned hybrid is near-symmetric with an insignificant bias, which implies that the model is not only able to estimate the sign but also the magnitude of the EEP without systematic over-or under pricing. Conversely, the Linear Regression error distribution exhibits fat tails- which is in line with the failure of the model to capture the non linear regression error distribution exhibits fat tails- which is in line with the failure of the model to capture the non linear EEP surface around the ATM options.

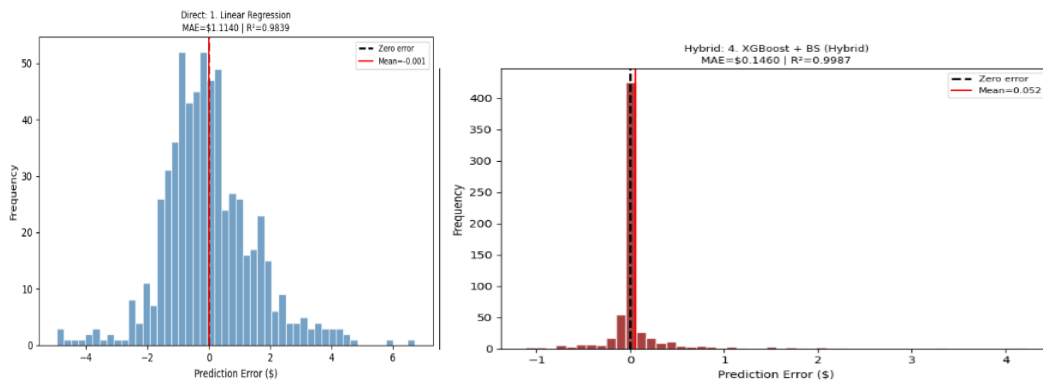


Fig. 3: Error distribution comparison between XGBoost Hybrid and Linear Regression

6.4 Bias-Variance Analysis

Comparison of training vs test MAE showed that XHBoost hybrid and the tuned models exhibit a small, yet, acceptable difference between training and test errors (gap < 0.10 per contract), suggesting good generalisation with minimal overfitting. The reason is that on random forest direct, the train-test gap is larger because of its ability to memorise training data at large max depth which is a typical high-variance behaviour. Linear Regression exhibits little train test gap, but great absolute errors on both sets- typical high bias underfittinh. The deep learning architectures (TensorFlow, PyTorch) have the advantage of Dropout regularisation (rate=0.10) and Batch Normalisation, which limit variance without significantly adding bias.

VII. Evaluation — Part 2: IV and Bid-Ask Spread Prediction

7.1 Implied Volatility Prediction

Table III shows the clean (leak-free) IV prediction results on the AAPL option chain, sorted by R^2 .

Table III: Implied Volatility Prediction — Clean Features (Sorted by R^2)

Model	MAE	MSE	RMSE	R^2
Random Forest	0.0817	0.1017	0.3189	0.5352
Gradient Boosting (GBM)	0.0946	0.1063	0.3260	0.5141
SVR (RBF)	0.0990	0.1265	0.3557	0.4217
Neural Network (MLP)	0.1220	0.1598	0.3998	0.2696
Linear Regression	0.1822	0.1489	0.3859	0.3193

Random Forest has the highest IV prediction results ($R^2 = 0.5352$, RMSE = 0.3189), then GBM ($R^2 = 0.5141$). The relatively low values of R^2 values in all models indicate the inherent difficulty of estimating IV using only contract

characteristics that can be observed. IV is a futuristic indicator of the market uncertainty that includes information absent in the option chain, especially the imbalance of the order flow, dealer hedging activity and expectations about macroeconomic news. These latent factors can be credited with the level of unexplained variance (about 47%).

The moneyness feature captures the systematic variation of IV with moneyness which is known as the volatility smile. The precise form of smile varies with the market conditions on a day to day basis forming a time varying behaviour that a fixed model when trained on a single day snapshot, cannot completely generalise. This is the reason why the best models still have a lot of unexplained variance in the prediction of IV.

7.2 Bid-Ask Spread Prediction

Table IV presents bid-ask spread prediction results, sorted by R^2 .

Table IV: Bid-Ask Spread Prediction — Clean Features (Sorted by R^2)

Model	MAE (\$)	MSE	RMSE (\$)	R^2
Neural Network (MLP)	0.2828	0.2455	0.4952	0.8988
Random Forest	0.2509	0.2494	0.4994	0.8971
GBM	0.3473	0.3142	0.5605	0.8703
SVR (RBF)	0.3328	0.3508	0.5923	0.8552
Linear Regression	1.1717	1.9174	1.3847	0.2087

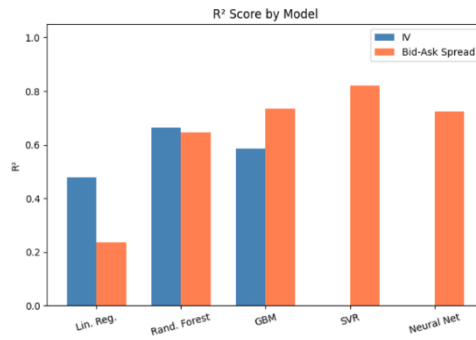


Fig. 4: Model performance comparison for IV and bid-ask spread prediction

The bid ask spread prediction shows a significantly higher value of R^2 than IV prediction in all models, where the neural network has $R^2 = 0.8988$ and the Random Forest has $R^2 = 0.8971$. This is in line with the theoretical assumption that spread is more directly related to visible liquidity measures- volume, open interest, and days to expiry- than IV, which reflects forward looking uncertainty.

Linear Regression in the case of spread prediction ($R^2 = 0.2087$) does badly compares with IV prediction, which is a very non-linear relationship between spread and its predictors. The non linear increase of the spread of very short dated options (gamma explosion), very deep OTM-options (low liquidity), and contracts with low open interest tree based model and neural network models are threshold effects that naturally behave non linearly in terms of their decision boundaries.

7.3 Market Microstructure Discussion

The outcomes of the bid ask spread are in the line with market microstructure theory of Ho and Stoll [11] and Glosten and Milgrom[18]. Spreads made by market makers in the AAPL options market are intended to cover: (1) the inventory risk – the cost of not hedging the delta; (2) the adverse selection risk – the risk of trading with knowledgeable counterparties with better information; and(3) the cost of processing the orders- the fixed operating cost per trade. High open interest and volume contracts have narrower spreads since market makers are less exposed to inventory risk (easy to hedge) and less exposed to adverse selection risk (more information on the market). These non- linear interactions among the liquidity proxies can be effectively learned in the Neural Network model.

VIII. Conclusions and Future Work

8.1 Summary of Findings

This article has proven that machine learning methods can greatly be used in the pricing of American options and in anticipating important variables in the market. In synthetic American option pricing, the hybrid EEP method — i.e. Black-Scholes as a structural prior, training ML models on the Early Exercise Premium residual, outperforms direct price prediction. Bayesian-tuned XG Boost Hybrid achieved near-perfect accuracy ($R^2 \approx 0.9999$, $MAE < \$0.05$), which proves the worth of the hybrid decomposition. There was similar performance between TensorFlow and PyTorch deep neural networks confirming the usefulness of deep learning models to options pricing.

In real-world AAPL option chain data, Random Forest was the strongest predictor of implied volatility ($R^2 = 0.5352$) while the Neural Network led on bid-ask spread estimation ($R^2 = 0.8988$). Linear Regression consistently gave inconsistent results in all activities, which offer empirical data that the variables of option markets are fundamentally non-linear and cannot be described by parametric linear models. The data leakage that has been detected and addressed in the first pipeline indicates that the careful selection of the features in financial use of ML is extremely important.

8.2 Limitations

There are number of limitations that must be mentioned. To start with, the synthetic dataset involves one fixed strike ($X=100$), which could constrain the model in generalising to different strike conditions. Second, the actual data were only gathered on one date and one underlying asset (AAPL), which limits the temporal and cross sectional predictability of IV and spread forecasts. Third, hyperparameter tuning was only done to SVR and XGboost for further optimisation of Bayesian optimisation to deep learning architectures (learning rate, dropout rate, architecture depth) may result in further improvements. Fourth, the hybrid methodology only supports American call options; to support American puts, there is a need to make changes to ensure that the various early exercise economics (interest rates as opposed to dividends) are taken into account.

8.3 Future Work

Some of the directions offer possibilities to research in future. Differential Machine Learning (DML), is an extension of the hybrid method suggested by by Huge and Savine [19], where the neural networks are trained on both prices and their Greeks ($\partial C/\partial S$, $\partial C/\partial \sigma$, and so on.), and the generalisation efficiency over limited training samples is substantially increased. An early exercise binary pre screening classifier might screen out contracts, compelling the EEP prediction to take the value of Zero on deep OTM options where early exercise does not optimize at all eliminating micro premium noise. Generalisation to stochastic volatility models (Heston [20]) and jump-diffusion models (Merton [21]) would create more realistic synthetic data. Lastly, the ability to collect realworld option chain data multivariate and multi underlying would also allow panel data models that would capture time dependent volatility smile processes that cannot be handled by single snapshot models.

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